



El and Cross Section

Cut the ski's layup at stations and compute, at each, the bending stiffness EI , torsional stiffness GJ , neutral axis and mass -- then the ski's flex numbers and weight.

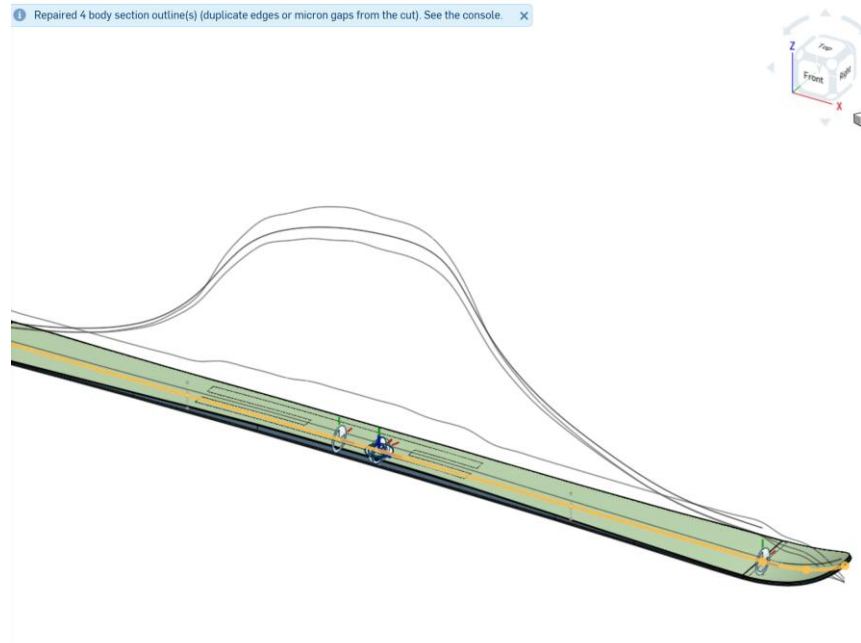


What it does

A ski's flex comes from its layup: every ply's shape and material. EI and Cross Section reads the layup straight from CAD (one solid per ply, each with an Onshape material), looks up the moduli in a CSV table, and gives $EI(x)$, $GJ(x)$, the neutral axis and the flex numbers -- so a design change shows its stiffness effect at once.

Use it when

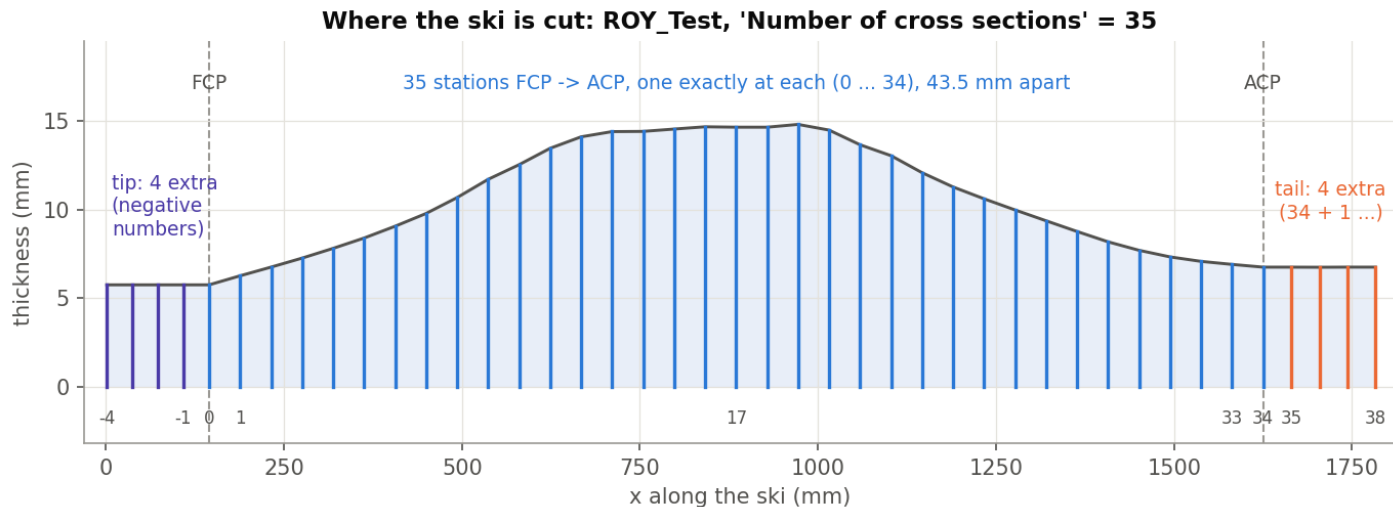
- You change a layup, core profile or sidecut
- You need the EI curve for deflection, baseline or profile work
- You compare a design with measured stiffness



ROY_Test: the output curves (EI , GJ ...) above the ski; the info reports repaired outlines.



Stations



FCP to ACP

N stations evenly spaced, one exactly at each contact point (numbered 0 ... N-1).

Tip and tail

Up to 4 extra each at about the same spacing; the tip's are numbered negative.

One cut per body

Outlines are cleaned (duplicates, micron gaps), triangulated and integrated.

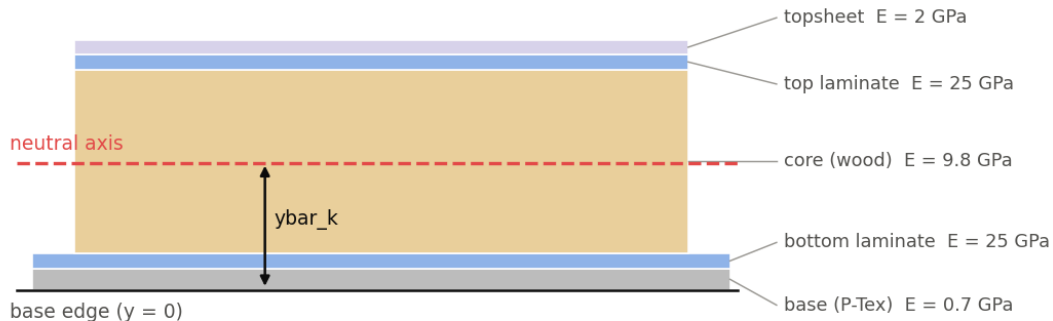


El of a laminate

Each body is one ply k : area A_k , centroid height y_{bar_k} , own I_k , modulus E_k

$$NA = \frac{\sum(E A y_{bar})}{\sum(E A)} \quad EI = \sum E (I_k + A_k y_{bar_k}^2) - \frac{(\sum E A y_{bar})^2}{\sum(E A)}$$

(the stiffness about the neutral axis; schematic section, thickness x 4)



Each body is a ply

Area, centroid height above the base, its own I , and its modulus E along the ski.

Neutral axis

The E -weighted centroid; EI is the stiffness about it.

E , not Q_{11}

A narrow ski bends like a beam: Q_{11} would overstate by $\sim 12\%$ (wood, Titanal), more for ± 45 fabric.



The dialog (1/2)

El and Cross Section 1 ✓ ✗

1

Wire to cross section along
El_Path ✕

2

Analysis name

3

FCP (Front Contact Point)
FCP ✕ 🕒

4

ACP (Aft Contact Point)
ACP ✕ 🕒

5

Cross section bodies
RD 21SKH 28 177 X-Sect ✕

6

Material library
📄 katai_ski_snowbaord_mater...

#	Parameter	What it means
1	Wire to cross section along	One wire along the ski (X increasing).
2	Analysis name	Prefix for the output curves.
3	FCP	Front contact point (vertex, planar face or mate connector; X only).
4	ACP	Aft contact point. FCP and ACP are optional, but flex numbers need both.
5	Cross section bodies	The layer solids (composites are unpacked).
6	Material library	materialData.csv: name, density, E, nu, Q11 ... Q66.



The dialog (2/2)

7 Refresh CSV data

8 Number of cross sections 35

9 ✓ Bodies & Materials

Bodies CLEAR

✓ RD 20ROY 28 178 4103... X
Body name RD 20ROY ...

✓ RD 20ROY 28 178 4103... X
Body name RD 20ROY ...

✓ RD 20ROY 28 178 4101... X
Body name RD 20ROY ...

✓ RD 20ROY 28 178 Tail... X
Body name RD 20ROY ...
Ignore (no stiffnes... ▾

✓ Part 7 (Poplar - Light) X
Body name Part 7

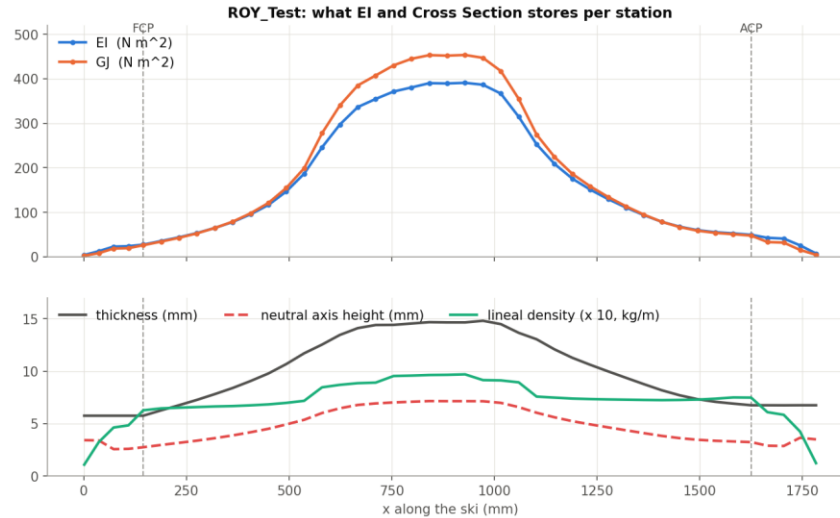
✓ Long_Beech (European... X
Body name Long_Beech

Progress bar: 0 to 100

#	Parameter	What it means
7	Refresh CSV data	Button: re-read the table.
8	Number of cross sections	3-200 (50); with FCP / ACP, this many from FCP to ACP.
9	Bodies & Materials	Per body: its material must match a CSV name exactly, or choose Ignore / Provide material data.
10	Output Options	Create composites (slow), language, material and body-detail tables.
11	Debug	Draw edges / points / mesh of chosen stations and bodies.



Example: ROY_Test, 35 stations

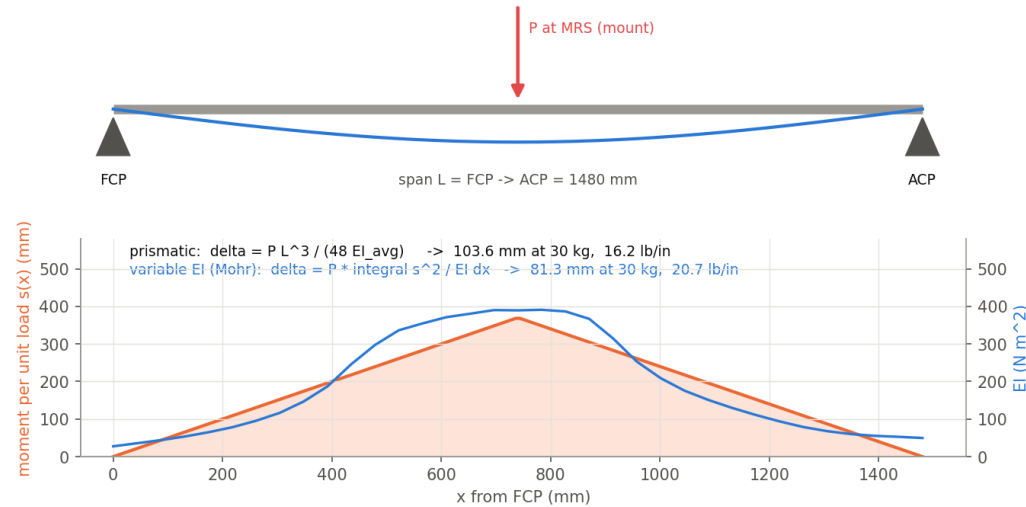


EI peaks at 390 N m² under the foot; GJ follows (453); the neutral axis sits near mid-thickness

Summary: 20.7 lb/in and 81.3 mm under 30 kg (variable EI); 16.2 lb/in prismatic. The stations cut 4 bodies with micron gaps, repaired (info).



Flex numbers



The test

Simply supported at FCP and ACP, a point load at the mount (MRS).

Prismatic

$P L^3 / (48 EI_{\text{avg}})$: the average EI over the span.

Variable EI (Mohr)

$P \times \int s^2 / EI$: the stiff middle counts more -- 20.7 vs 16.2 lb/in on ROY_Test.



Outputs and messages

Outputs

Key	What it holds
<name>_EI	EI curve: $Z \text{ mm} = EI \text{ N m}^2$. The input of every other xSection feature.
_GJ, _neutralAxis, _linealDensity, _profileHeight	GJ ($\text{mm} = \text{N m}^2$), NA height, mass per length ($1 \text{ kg/m} = 100 \text{ mm}$), thickness (true scale).
CrossSectionAnalysis (attribute)	Everything per station, read by the Cross-Section Analysis table and Update profile.

Messages

	Message / what to do
Info	Repaired N body section outline(s)... Duplicate edges or micron gaps closed; details on the console.
Warning	N body section(s) between FCP and ACP could not be closed or came out with no area EI is low there.



Tips and limits



Name materials exactly

The Onshape material name must match the CSV Name, case and spaces included; otherwise the body is ignored or needs data.



Turn off Create composites

It is the slowest step and only draws the section wires.



Wood cores

Some CSV rows (Aspen, Poplar) are isotropic placeholders: EI is fine, their GJ is overstated.



Ends

Planes grazing the tip or tail can lose a thin outline (console only).



Where it lives

Document	xSection (f8deedeb1fbd819a8fa20113), 'Stiffness Tools'
Tab	xSect (features/xSect.fs + core/, section/, materials/, beam/)
Tests	xSection tests studio: analytic EI / NA / GJ (check_xsection_tests.py)
Examples	ROY_Test
Explainer	docs/explainers/xsection/xsection_explained.md, section 1.1-1.6, 2.1
Related	EI and Cross Section, Estimate Stiffness, Estimate Deflection, Generate / Analyze baseline, Update profile